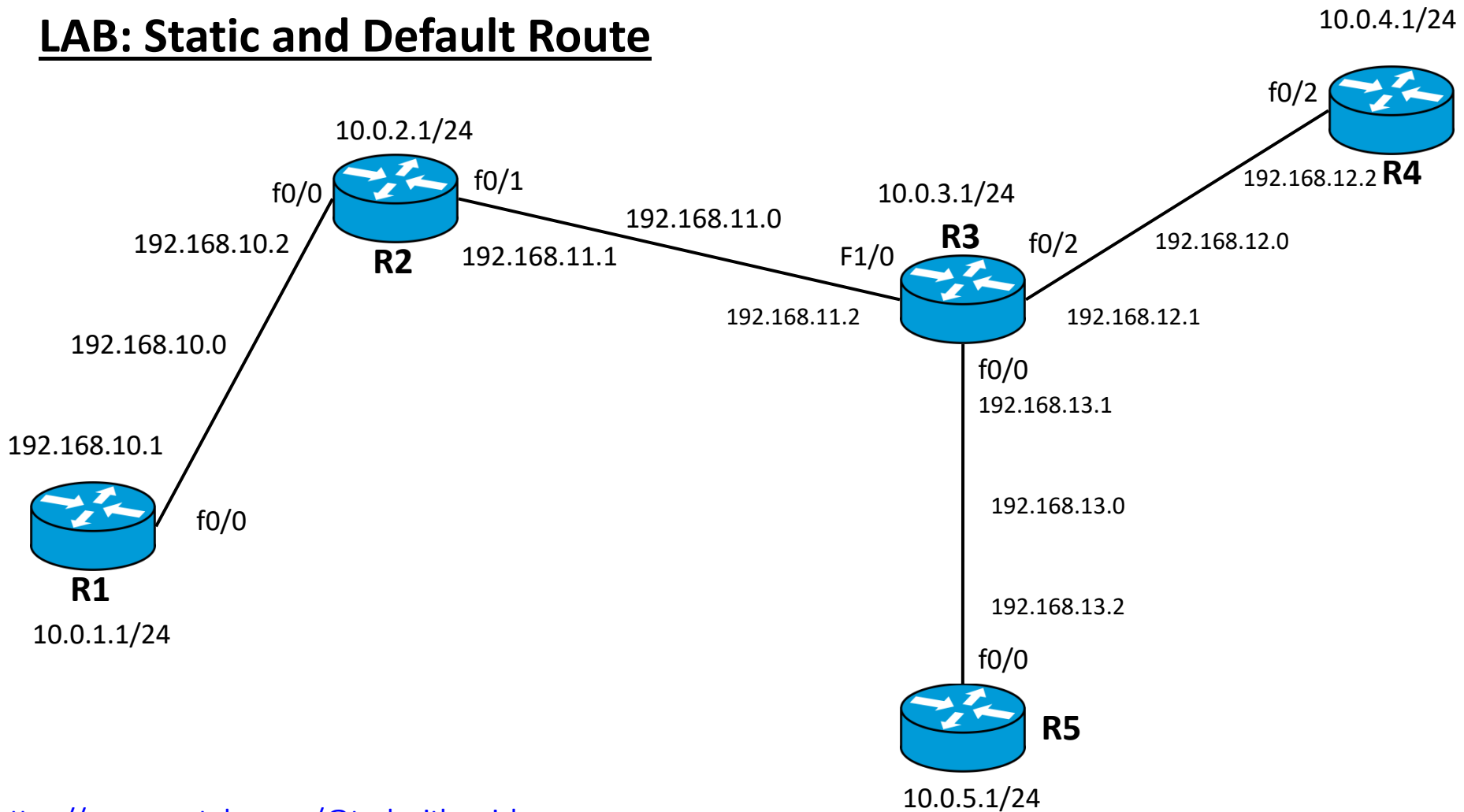


LAB: Static and Default Route



STEP 1: Find all Indirect Networks and next hop of Routers

R1: Indirect networks of R1 and next hop Addresses

10.0.2.0 with next hop 192.168.10.2

10.0.3.0 with next hop 192.168.10.2

10.0.4.0 with next hop 192.168.10.2

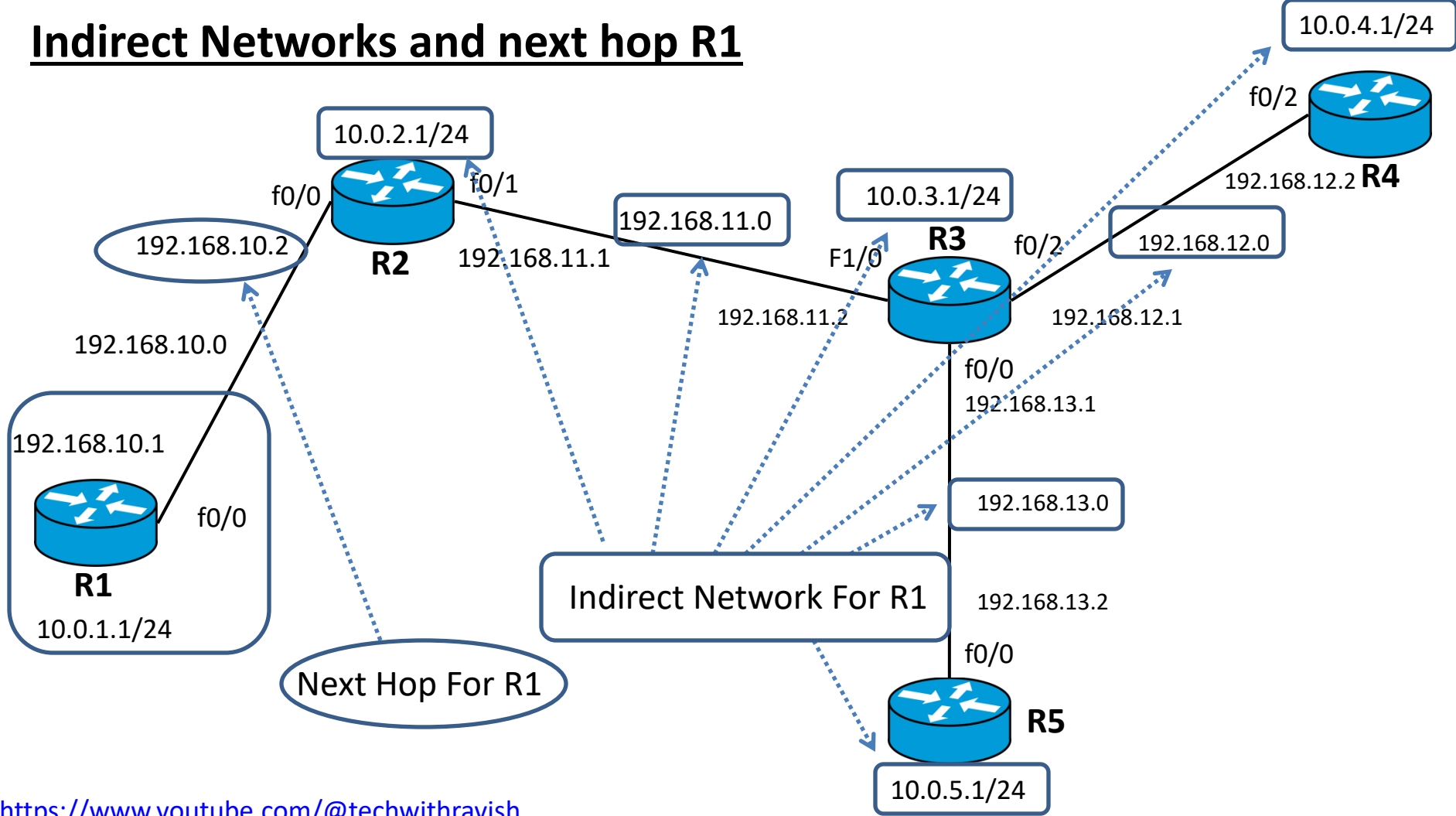
10.0.5.0 with next hop 192.168.10.2

192.168.11.0 with next hop 192.168.10.2

192.168.12.0 with next hop 192.168.10.2

192.168.13.0 with next hop 192.168.10.2

Indirect Networks and next hop R1



R2: Indirect networks of R2 and next hop Addresses

10.0.1.0 with next hop 192.168.10.1

10.0.3.0 with next hop 192.168.11.2

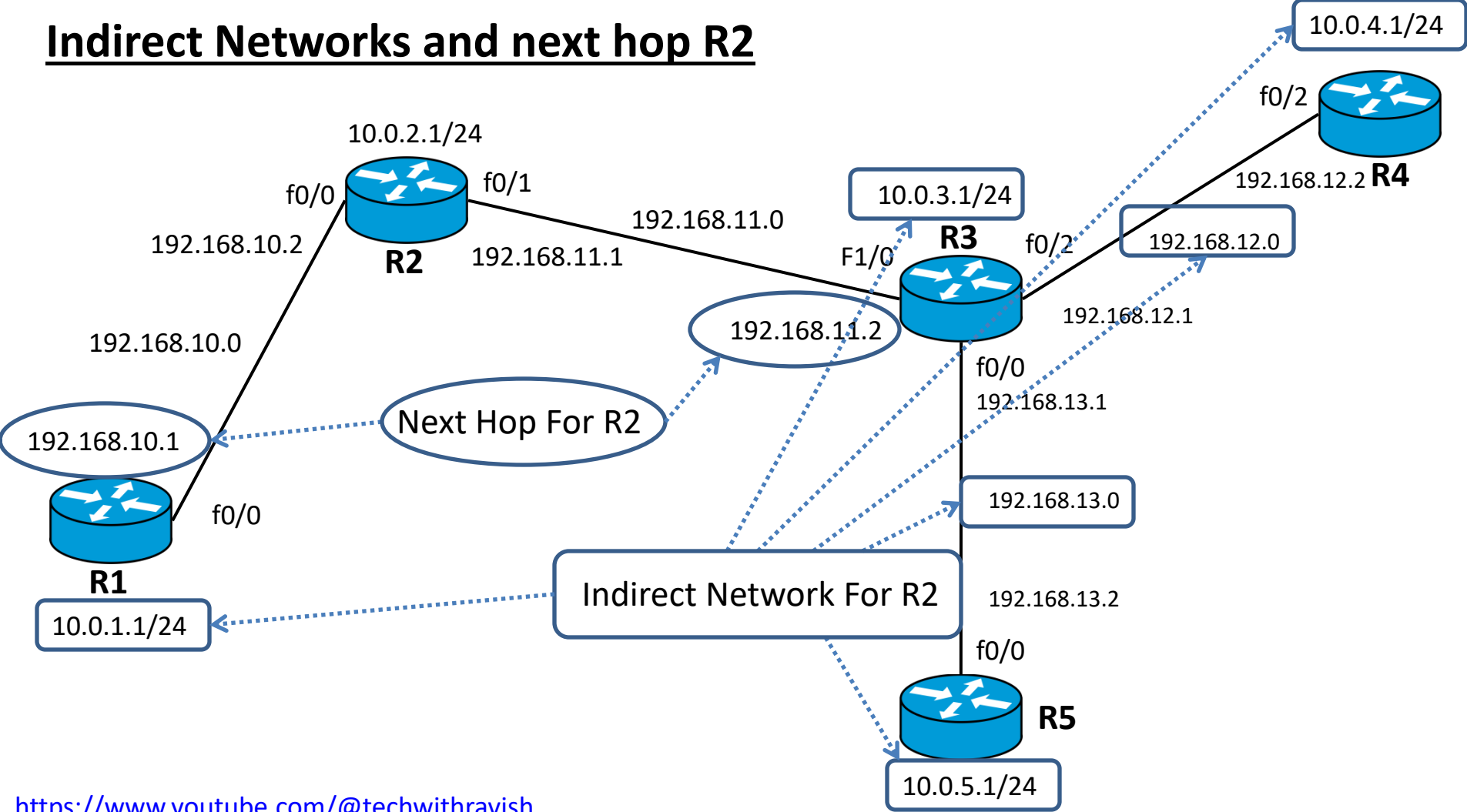
10.0.4.0 with next hop 192.168.11.2

10.0.5.0 with next hop 192.168.11.2

192.168.12.0 with next hop 192.168.11.2

192.168.13.0 with next hop 192.168.11.2

Indirect Networks and next hop R2



R3: Indirect networks of R3 and next hop Addresses

10.0.1.0 with next hop 192.168.11.1

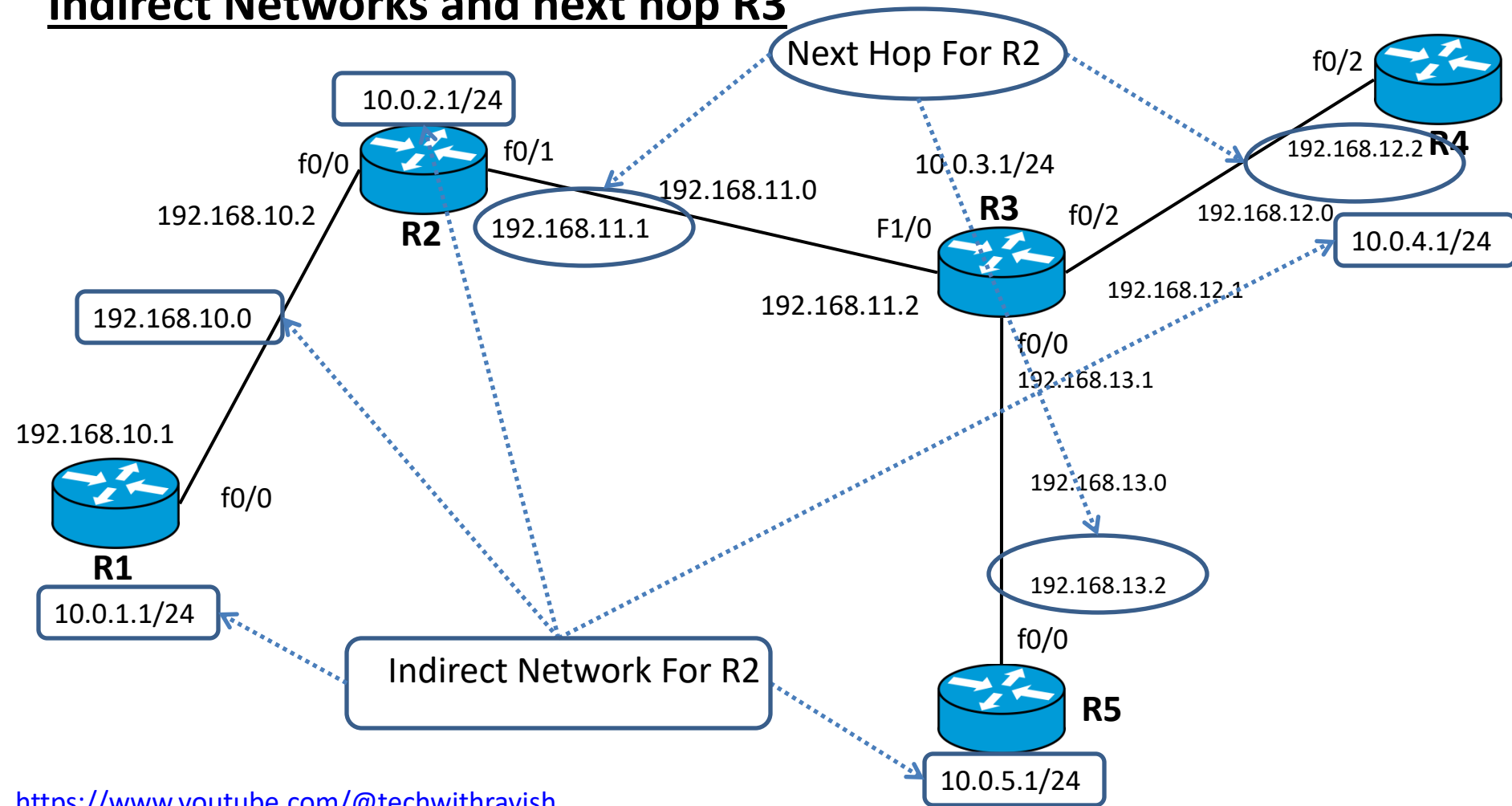
10.0.2.0 with next hop 192.168.11.1

10.0.4.0 with next hop 192.168.12.2

10.0.5.0 with next hop 192.168.13.2

192.168.10.0 with next hop 192.168.11.1

Indirect Networks and next hop R3



R4: Indirect networks of R4 and next hop Addresses

10.0.1.0 with next hop 192.168.12.1

10.0.2.0 with next hop 192.168.12.1

10.0.3.0 with next hop 192.168.12.1

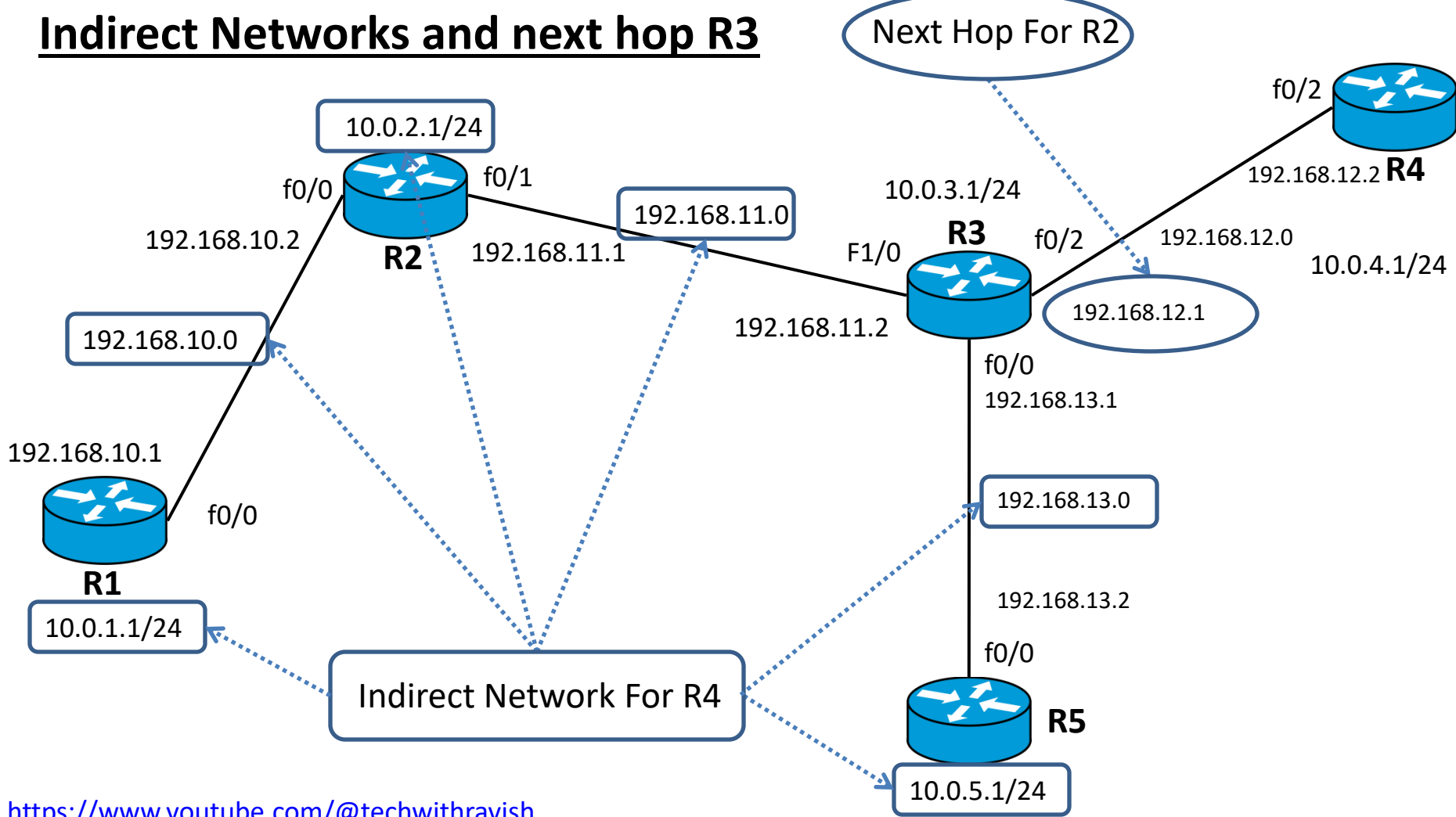
10.0.5.0 with next hop 192.168.12.1

192.168.10.0 with next hop 192.168.12.1

192.168.11.0 with next hop 192.168.12.1

192.168.13.0 with next hop 192.168.12.1

Indirect Networks and next hop R3



R5: Indirect networks of R5 and next hop Addresses

10.0.1.0 with next hop 192.168.13.1

10.0.2.0 with next hop 192.168.13.1

10.0.3.0 with next hop 192.168.13.1

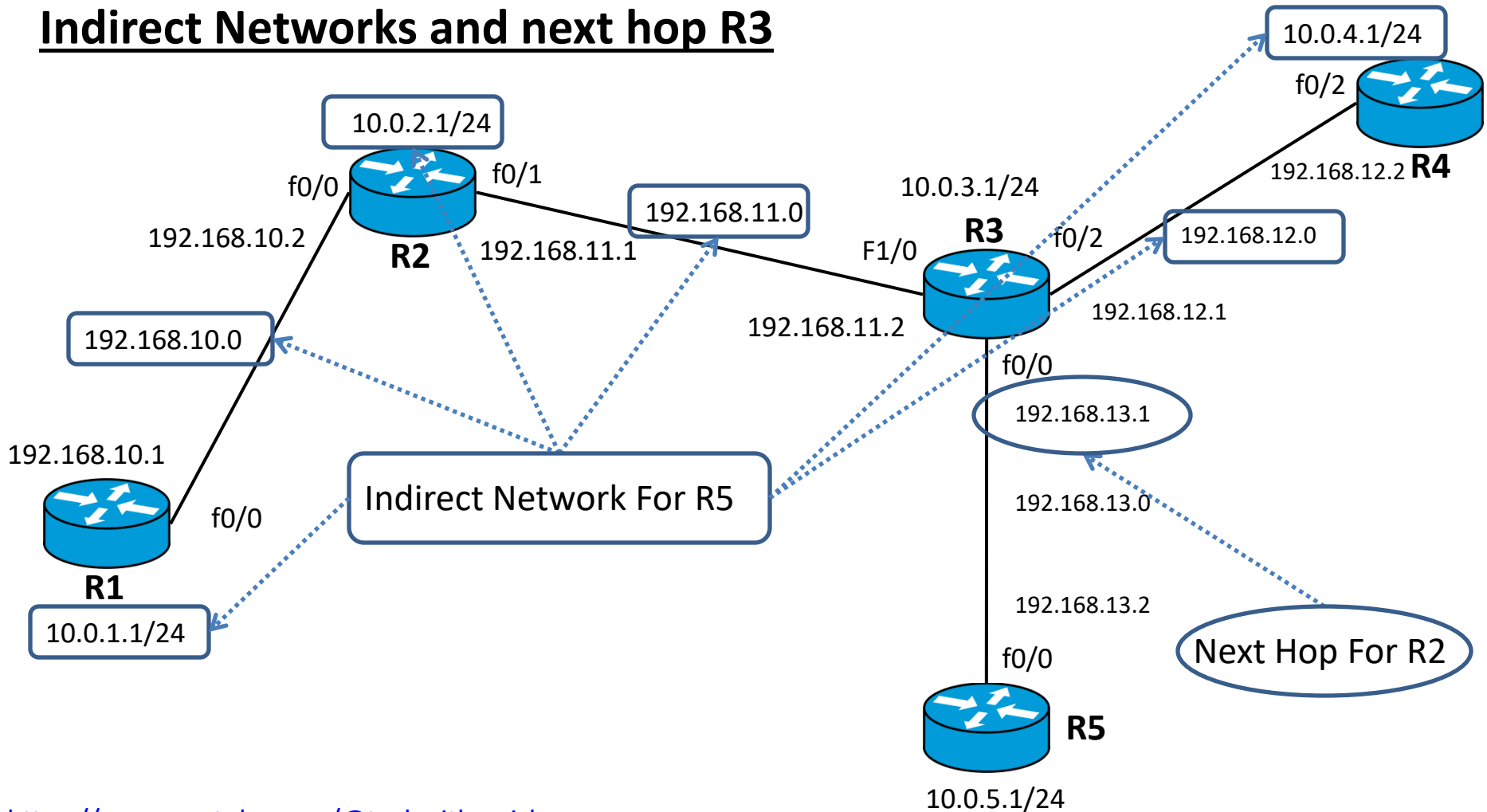
10.0.4.0 with next hop 192.168.13.1

192.168.10.0 with next hop 192.168.13.1

192.168.11.0 with next hop 192.168.13.1

192.168.12.0 with next hop 192.168.13.1

Indirect Networks and next hop R3



Configure IP on Router 1:

R1#conf t

R1(config)#int loopback 1

R1(config-if)#ip address 10.0.1.1 255.255.255.0

R1(config-if)#Exit

R1(config)#int f0/0

R1(config-if)#no shutdown

R1(config-if)#ip address 192.168.1.1 255.255.255.0

R1(config-if)#exit

Configure IP on Router 2:

```
R2#conf t
```

```
R2(config)#int loopback 1
```

```
R2(config-if)#ip address 10.0.2.1 255.255.255.0
```

```
R2(config-if)#exit
```

```
R2(config)#int f0/0
```

```
R2(config-if)#no shutdown
```

```
R2(config-if)#ip address 192.168.10.2 255.255.255.0
```

```
R2(config-if)#ex
```

```
R2(config)#int f1/0
```

```
R2(config-if)#no shutdown
```

```
R2(config-if)#ip address 192.168.11.1 255.255.255.0
```

```
R2(config-if)#exit
```

Configure IP on Router 3:

R3#

R3#conf t

R3(config)#int loopback 1

R3(config-if)#ip address 10.0.3.1 255.255.255.0

R3(config-if)#exit

R3(config)#int f2/0

R3(config-if)#no shutdown

R3(config-if)#ip address 192.168.12.1 255.255.255.0

R3(config-if)#exit

R3(config)#int f0/0

R3(config-if)#no shutdown

R3(config-if)#ip address 192.168.13.1 255.255.255.0

R3(config-if)#exit

R3(config)#int f1/0

R3(config-if)#no shutdown

R3(config-if)#ip address 192.168.11.2 255.255.255.0

R3(config-if)#exit

Configure IP on Router 4:

```
R4(config)#int f0/0
```

```
R4(config-if)#no shutdown
```

```
R4(config-if)#ip address 10.0.4.1 255.255.255.0
```

```
R4(config-if)#exit
```

```
R4(config)#int f2/0
```

```
R4(config-if)#no shutdown
```

```
R4(config-if)#ip address 192.168.12.2 255.255.255.0
```

```
R4(config-if)#exit
```

Configure IP on Router 5:

```
R5#conf t
```

```
R5(config)#int f0/0
```

```
R5(config-if)#no shutdown
```

```
R5(config-if)#ip address 192.168.13.2 255.255.255.0
```

```
R5(config-if)#exit
```

```
R5(config)#int loopback 1
```

```
R5(config-if)#ip address 10.0.5.1 255.255.255.0
```

```
R5(config-if)#exit
```


Configure Static Routing on Router 1:

R1#conf t

R1(config)#ip route 10.0.2.0 255.255.255.0 192.168.10.2

R1(config)#ip route 10.0.3.0 255.255.255.0 192.168.10.2

R1(config)#ip route 10.0.4.0 255.255.255.0 192.168.10.2

R1(config)#ip route 10.0.5.0 255.255.255.0 192.168.10.2

R1(config)#ip route 192.168.11.0 255.255.255.0 192.168.10.2

R1(config)#ip route 192.168.12.0 255.255.255.0 192.168.10.2

R1(config)#ip route 192.168.13.0 255.255.255.0 192.168.10.2

Configure Static Routing on Router 2:

```
R2#conf t
```

```
R2(config)#ip route 10.0.1.0 255.255.255.0 192.168.10.1
```

```
R2(config)#ip route 10.0.3.0 255.255.255.0 192.168.11.2
```

```
R2(config)#ip route 10.0.4.0 255.255.255.0 192.168.11.2
```

```
R2(config)#ip route 10.0.5.0 255.255.255.0 192.168.11.2
```

```
R2(config)#ip route 192.168.12.0 255.255.255.0 192.168.11.2
```

```
R2(config)#ip route 192.168.13.0 255.255.255.0 192.168.11.2
```

Configure Static Routing on Router 3:

```
R3#conf t
```

```
R3(config)#ip route 10.0.1.0 255.255.255.0 192.168.11.1
```

```
R3(config)#ip route 10.0.2.0 255.255.255.0 192.168.11.1
```

```
R3(config)#ip route 10.0.4.0 255.255.255.0 192.168.12.2
```

```
R3(config)#ip route 10.0.5.0 255.255.255.0 192.168.13.2
```

```
R3(config)#ip route 192.168.10.0 255.255.255.0 192.168.11.1
```

Configure Static Routing on Router 4:

```
R4#conf t
```

```
R4(config)#ip route 10.0.1.0 255.255.255.0 192.168.12.1
```

```
R4(config)#ip route 10.0.2.0 255.255.255.0 192.168.12.1
```

```
R4(config)#ip route 10.0.3.0 255.255.255.0 192.168.12.1
```

```
R4(config)#ip route 10.0.5.0 255.255.255.0 192.168.12.1
```

```
R3(config)#ip route 192.168.10.0 255.255.255.0 192.168.12.1
```

```
R3(config)#ip route 192.168.11.0 255.255.255.0 192.168.12.1
```

```
R3(config)#ip route 192.168.13.0 255.255.255.0 192.168.12.1
```

Configure Static Routing on Router 5:

```
R5#conf t
```

```
R5(config)#ip route 10.0.1.0 255.255.255.0 192.168.13.1
```

```
R5(config)#ip route 10.0.2.0 255.255.255.0 192.168.13.1
```

```
R5(config)#ip route 10.0.3.0 255.255.255.0 192.168.13.1
```

```
R5(config)#ip route 10.0.4.0 255.255.255.0 192.168.13.1
```

```
R5(config)#ip route 192.168.10.0 255.255.255.0 192.168.13.1
```

```
R5(config)#ip route 192.168.11.0 255.255.255.0 192.168.13.1
```

```
R5(config)#ip route 192.168.12.0 255.255.255.0 192.168.13.1
```

Alternative to Static Route:

On end point Routers we can do Default Route Configuration

**Endpoint routers are those routers which has a single next hop:
Like R1, R4 and R5**

```
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.10.2
```

```
R4(config)#ip route 0.0.0.0 0.0.0.0 192.168.12.1
```

```
R5(config)#ip route 0.0.0.0 0.0.0.0 192.168.13.1
```